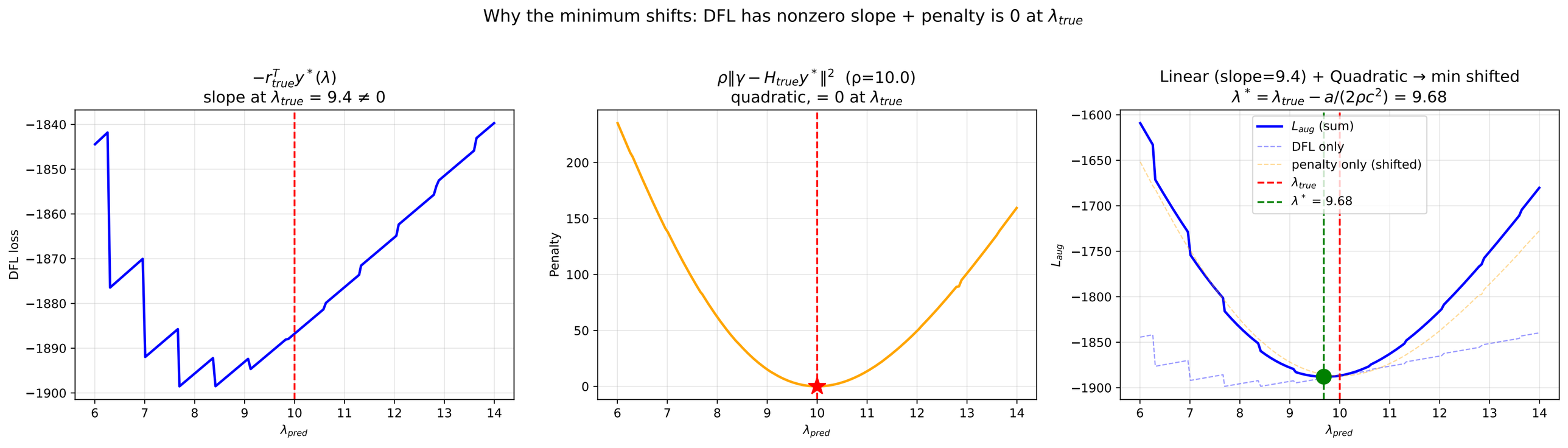
Why the minimum shifts: DFL has nonzero slope + penalty is 0 at λ_{true}
$-r_{true}^T y^*(\lambda)$
slope at λ_{true} = 9.4 ≠ 0
DFL loss
λ_{pred}
$\rho\|\gamma - H_{true} y^*\|^2$ (ρ=10.0)
quadratic, = 0 at λ_{true}
Penalty
λ_{pred}
Linear (slope=9.4) + Quadratic → min shifted
$\lambda^* = \lambda_{true} - a/(2\rho c^2)$ = 9.68
L_{aug}
L_{aug} (sum)
DFL only
penalty only (shifted)
λ_{true}
λ^* = 9.68
λ_{pred}